K223: Nuclear $\gamma\gamma$ Angular Correlations

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1 Introduction

The aim of this experiment is to study the angular distribution of the gamma raditaion from the $\gamma - \gamma$ cascade from the decay of ^{60}Co to ^{60}Ni, by setting up a Fast-Slow Coincidence circuit. After first calibrating the components of the circuit setup, we set one detector to rotate in a 180° range around the ^{60}Co source, while a second detector sat stationary at the same distance from the source as the first outside the semicircle traced out by the first detector. With this setup we took first a measurement of the random coincidences, followed by a measurement of the true coincidences.

2 Theory

2.1 *Circuit Components*

2.1.1 *Scintillator Detector*

A scintillator detector consists of a photomultiplier optically coupled to a scinitllating material (Fig. 1). The photomultiplier is a tube containing a photocathode, a row of dynodes, and an anode (Fig. 2). When a high voltage is applied to this structure, a potential 'ladder' is formed from the cathode to the dynodes to the anode. Thus, when a photon strikes the cathode, the electron that is emitted goes toward the first dynode, where it transfers its energy and causes secondary electrons to be emitted, which go towards the second dynode, and so on. This process results in a cascade of electrons reaching the anode at the end, which creates an electric current proportional to the energy (and therefore wavelength) of the photon that started this process.

The scintillating material when struck by a photon, absorbs it then re-emits another photon with a different wavelength. In this experiment, the scintillator we used

was one of NaI crystals with Tl doping. In inorganic crystals like this, the scintillation mechanism is a characteristic of their electronic band structure. While an incident photon would normally excite an electron from the valence band to the conduction band (if it is of the correct wavelength), the excited electron can only fall back down to the valence band, emitting a photon of the same wavelength. The presence of impurities in the crystal allow for the existence of an exciton band, an energy band just below the conduction band into which electrons can also be excited. This allows for a range of possible frequencies of photons that can be emitted and excited.

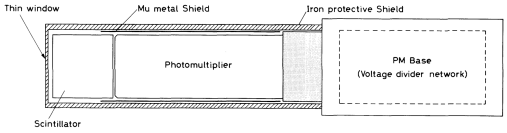


Figure 1: Schematic diagram of a scintillator [1]

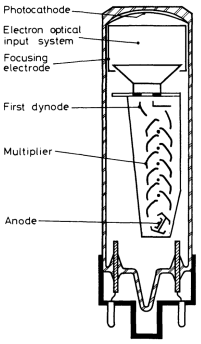


Figure 2: Schematic diagram of a photomultiplier [1]

2.1.2 *Single Channel Analyser (SCA)*

An SCA takes in an analog signal and outputs a digital square wave signal, which depends on the amplitude of the input signal. It has an upper and a lower amplitude threshold which can both be adjusted, and any signals falling outside these are blocked.

2.1.3 *Multichannel Analyser (MCA)*

Like an SCA, an MCA too takes in analog signals and converts them to a digital output. However, as the name suggests, it is capable of outputting a larger range of digital values (instead of just 1 and 0 like an SCA). It thus has a series of thresholds, like the bins of a histogram, with signals falling into different 'bins' producing different digital responses. The MCA also stores these outputs in channel bins reflecting the thresholds, which thus can be used to generate a spectrum showing the different amplitudes it has received as input.

While the thresholds of an MCA are set by the manufacturers and cannot be modified like those of an SCA, we can input into an MCA a square wave gate signal, acting as thresholds outside of which the MCA will not record any input signals.

2.1.4 *Constant Fraction Discriminator (CFD)*

Two coincident signals with different amplitudes can cause a walk effect in an SCA, where they cross its lower threshold at different times resulting in different, time-delayed outputs for the two. One way to correct this is to use a CFD, which instead of producing a digital signal at the moment an input analog signal crosses a threshold, generates the output at a constant fraction of the peak height, which eliminates any walk effects.

The way the CFD works is by splitting an incoming pulse into two. One part is delayed by the time needed for the signal to rise from the chosen constant fraction level to its peak, while the other part is inverted and attenuated by the constant fraction. The two parts are then added together, and the point where the resulting signal has zero amplitude is then at the chosen constant fraction of the signal height.

2.2 $\gamma - \gamma$ *cascade and angular correlation*

For a naïve treatment of the directional correlation from a $\gamma - \gamma$ cascade, we begin by first considering a single γ-transition. Let this transition (of angular momentum $\mathbf{L}$) be from an initial nuclear level i to a final nuclear level f, with their spins being $\mathbf{I}_i$ and $\mathbf{I}_f = \mathbf{I}_i - \mathbf{L}$ respectively. The quantum numbers of these two levels are thus I_i, m_i and I_f, m_f respectively. The gamma ray has an angular momentum quantum number (or multipolarity) L and a magnetic quantum number $M = m_i - m_f$.

For every γ-transition between some m_i to some m_f, there is a directional distribution $F_L^M(\theta)$ (θ being the angle between the emitted gamma ray and the Z-axis) which is independent of the spins $\mathbf{I}_i$ and $\mathbf{I}_f$. Since we are generally unable to observe the splitting of the radiation from these different components (as in the Zeeman effect), we generally observe the unresolved line, containing the transitions between every possible pair of magnetic quantum numbers. This has a directional distribution $F_L(\theta)$, to calculate which we need all the $F_L^M(\theta)$, the relative population $P(m_i)$ of each sub-level m_i, and the relative transition probability $G(m_i, m_f)$ for each pair of m_i and m_f. Then,

$$F_L(\theta) \propto \sum_{m_i, m_f} P(m_i)\ G(m_i, m_f)\ F_L^M(\theta)$$

Using the Clebsch-Gordan coefficients for the quantum numbers I_i, m_i, I_f, m_f allows us to replace the relative transition probability as

$$G(m_i, m_f) = |\langle I_f m_f L M | I_i m_i \rangle|^2$$

We can now consider a $\gamma - \gamma$ cascade which we will assume happens through the nuclear levels $I_i \to I \to I_f$, and emits two gamma-rays of multipolarity L_1 and L_2. We can set our coordinate axis such that the Z-axis is in the direction of the first

gamma-ray's emission. Then, the directional correlation function $W(\theta)$ of the cascade is simply the distribution $F_L(\theta)$ of the second gamma-ray with respect to the Z-axis.

The distribution can thus be calculated once each relative probability $P(m)$ for the intermediate sub-levels is known. This is given by the sum of all transitions $m_i \rightarrow m$. Assuming all m_i states are equally populated, we have

$$P(m) \propto \sum_{m_i} G(m_i, m)\ F_{L_1}^{M_1}(\theta = 0)$$

A propagating photon can only carry in its direction of motion an angular momentum $\pm h$, and because we had chosen the Z-axis to be along the direction of the first gamma-ray, we can restrict M_1 to the values ± 1. So, the directional correlation can be calculated as

$$\begin{aligned} W(\theta) = F_L(\theta) &\propto \sum_{m,m_f} P(m)\ G(m, m_f)\ F_{L_2}^{M_2}(\theta) \\ \implies W(\theta) &\propto \sum_{m_i,m,m_f} |\langle ImL_1 \pm 1 | I_i m_i \rangle|^2\ F_{L_1}^{\pm 1}(0)\ |\langle I_f m_f L_2 M_2 | I_i m_i \rangle|^2\ F_{L_2}^{M_2}(\theta) \end{aligned}$$

When the gamma-rays emitted in the cascade are both pure (i.e. L_1 and L_2 have one value each instead of being of a mixed multipole character), the above expression can be simplified to

$$W(\theta) = 1 + A_{22} P_2(\cos\theta) + \ldots + A_{k_{max} k_{max}} P_{k_{max}}(\cos\theta)$$

where P_n are the Legendre polynomials, and

$$k_{max} = \min(2I, 2L_1, 2L_2)$$

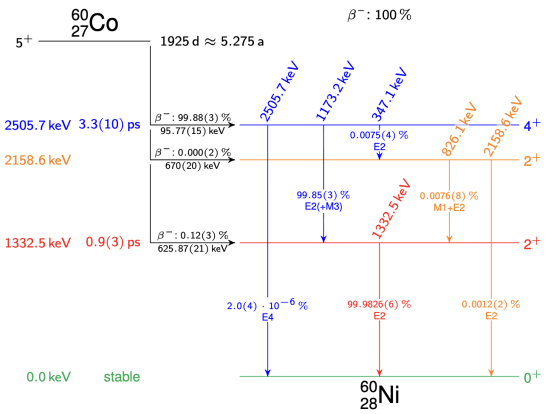


Figure 3: Decay Scheme of ^{60}Co [2]

As can be seen from the decay scheme in Fig. 3, ^{60}Co decays via a $\gamma - \gamma$ cascade into the stable ^{60}Ni, which has 28 protons and 32 neutrons. (Though there are multiple ways in which the decay occurs, the transitions from the 2505.7 keV to the 1332.5 keV and then to the ground state are most predominant.) Such an 'even-even' nucleus is useful for angular correlation observations of the cascade. The nucleus presumably has a spin 0 in its ground state and hence the second γ-transition must be pure. Only the first γ-transition can be mixed.

Experimentally, this $\gamma - \gamma$ cascade has been shown to occur between spins 4, 2, and 0, and the corresponding directional correlational function has been calculated to be of the form

$$W(\theta) = 1 + 0.1020 P_2(\cos\theta) + 0.0091 P_4(\cos\theta)$$

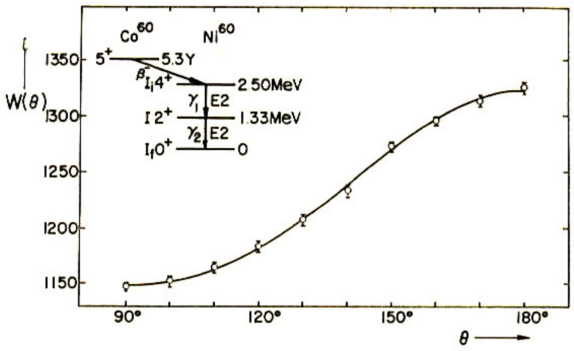


Figure 4: Experimentally obtained directional correlation distribution of the ^{60}Co to ^{60}Ni cascade[3]. The solid line is the theoretically determined function $W(\theta)$ [4]

3 Preparatory tasks

3.1 *Which distances to use?*

How does the count rate of one detector change with the distance?
The count rate of a detector follows the inverse square law with respect to distance, i.e., it drops by a factor of d^{-2} as the distance d from the detector increases.

How does the coincidence rate change with the distance? Give a formula!
Since coincidences require both detectors to register radiation, the coincidence rate is related to the distance of both detectors as $f \propto d_1^{-2} d_2^{-2}$.

How does the asymmetry seen in the experiment differ from the theoretical prediction? Again, give a formula.
The asymmetry in the experiment arises due to the presence of noise, which we can take to be Gaussian in nature. The probability distribution for Gaussian noise is

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-x^2}{2\sigma^2}}$$

where σ is the variance of the noise.

How do the correction factors influence the error of the corrected correlation coefficients?
As the correlation coefficients are scaled down by the correction factors, the error in them is also scaled down by this factor.

Using the values provided in Siegbahn for 2 "× 2 " crystals: which distance is likely to yield most precise results?
According to the data in Siegbahn, keeping the detectors 10 cm away from the source

is most likely to yield precise results. This is simply due to the smaller solid angle the detector takes up by being further away. In an ideal case, the angular correlation measurement would occur at a single point, hence reducing the solid angle improves the values we measure.

3.2 *Which angles to pick?*

The theoretical prediction for the angular correlation of 4-2-0 cascade is

$$f(\theta) = N(1 + A_{22}P_2(\cos^2\theta) + A_{44}P_4(\cos^4\theta)).$$

With Legendre polynomials

$$P_2(x) = \frac{1}{2}(3x^2 - 1) \text{ and } P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3)$$

this can be written as

$$f(\theta) = A(1 + B\cos^2\theta + C\cos^4\theta)$$

with

$$A = N(1 + \frac{3}{8}A_{44} - \frac{1}{2}A_{22}) \tag{1}$$

$$B = \frac{A_{22}\frac{3}{2} - A_{44}\frac{30}{8}}{1 + \frac{3}{8}A_{44} - \frac{1}{2}A_{22}} \tag{2}$$

$$C = \frac{\frac{35}{8}A_{44}}{1 + \frac{3}{8}A_{44} - \frac{1}{2}A_{22}} \tag{3}$$

$$\tag{4}$$

where A is a scaling factor and B and C are the parameters we're interested in. By changing parameters instead to $\alpha = B + C$, $\beta = B - C$, the above function changes to

$$f(\theta) = A\left(1 + \left(\frac{\alpha + \beta}{2}\right)\cos^2\theta + \left(\frac{\alpha - \beta}{2}\right)\cos^4\theta\right).$$

The function is plotted agains the angle and is shown in Figure 5.

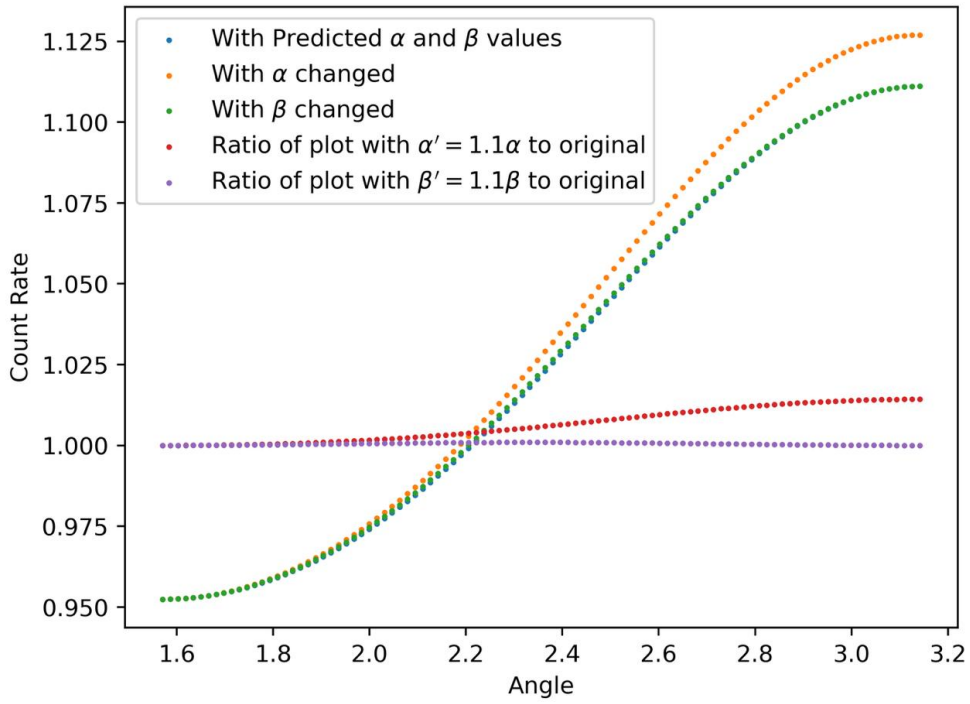


Figure 5: Angular Correlation Function and their Variation

Here α and β are varied by 10 %. The ratios of varied and original values is also plotted to see the major differences. Change in α has here the most impact, whereas β has close to no influence on the function. It is seen that around 180° the function experiences the most change. Therefore the measured angles should range from approximately 120° to 240°. Here the constants $A_{22} = 0.102$ and $A_{44} = 0.0091$ are chosen since the $4-2-0$ cascade is expected. Since at the end of the expereiment also other cascades are compared to the measurements, the measured angle range is still chosen to be from 90° to 270° in 15° steps, totaling to 13 angles.

3.3 *How to correct for de-adjustment?*

Let the source be at distance of k away from the centre of the circle in the direction of the stationary detector. We can determine the new distance between the source and the moving director at an angle θ using the Law of Cosines (where θ is the angle between the lines formed by joining the centre of the circle to the two detectors). If the original distance, the radius of the circle was r, the new distance is

$$d = \sqrt{r^2 + k^2 - 2rk\cos\theta}$$

. Since for a circle of radius 1, the count rate would be $f(\theta) = 1$, and since the count rate follows the inverse square law with distance from the source, we know the count rate originally was

$$f(\theta) = \frac{1}{r^2}$$

which becomes now

$$f(\theta) = \frac{1}{d^2} = \frac{1}{r^2 + k^2 - 2rk\cos\theta}$$

This will appear in the data as a high spike near $\theta = 0°$ before sharply falling and plateauing towards 0 at $\theta = 180°$. The larger the deviation from the centre, the sharper the fall and the closer the latter part of the data will be to 0.

4 Preliminary Set Up

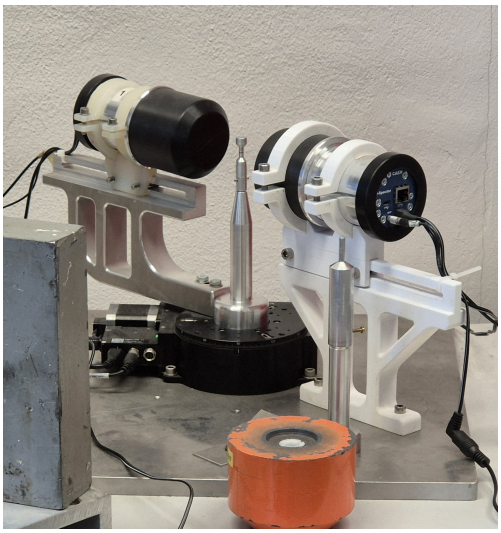


Figure 6: Image of the Experimental Setup. The two detectors sit on opposite sides of the ^{60}Co source. The detector nearer to the camera was kept stationary in the experiment, while the one further away was rotated; at the moment (and for the calibration part of the experiment) the angle between them is 180°. The orange container in the front is the container for the radioactive source.

At the beginning of the experiment, the movable detector was set 180° across from the stationary detector, directly opposite it. It was kept in this position for the first part of the experiment (the steps detailed in the next subsection).

The first thing we did was measure the distance of the source from the two detectors. Since the source was placed in between the two, we did this by measuring the

distance between the two detectors and halving it. This distance was measured to be 90 ± 1 mm, which halves to give us a distance between the source and the detectors of 45 ± 0.5 mm. However, both detectors had a cap of thickness 5 mm, meaning the actual distance between the source and each detector was 50 ± 0.5 mm.

5 Preparing and Calibrating the Setup

5.1 *Setting Gain of Main Amplifiers*

When a signal fed into the MCA is amplified, its voltage increases, and the MCA sees smaller and smaller fractions of the original signal, that part whose voltage falls within its range. Effectively, we are zooming in on a section of the MCA, seeing the signal with greater resolution. Therefore, it is ideal for us to be having as high an amplification to our signal as possible, without having the features of interest of the signal go out of frame for the MCA.

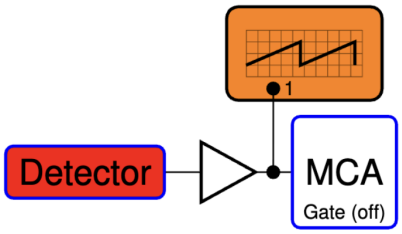


Figure 7: Circuit Diagram for setting amplifier gain. The triangle represents the amplifier, while the orange box is the oscilloscope which here is receiving a signal in channel 1. [5]

In the experiment, we went about this by manually increasing the amplitude on the two main amplifiers until the visible photopeaks from the signals from both the moving and stationary detector were at the right edge of the MCA screen. Since no external gate signal was being sent to the MCA (with its input being set to 'Spectrum' and 'Gate' being set to 'Off'), we saw the full spectrum of the analog signal. However, before we could increase the amplification to the point where the photopeaks were on the right edge of the MCA plot, the signal began getting saturated (the sine curve plateauing). Hence, we only amplified the signal to just before this occurred, keeping the photopeaks in the MCA screen around the centre of the plots.



Figure 8: Signal from the moving detector on the oscilloscope



Figure 9: Signal from the stationary detector on the oscilloscope

In the MCA plot of the full spectrum of the signal, we identified two prominent narrow peaks that were close to each other (just before channel 6000 in Fig. 10) as the peaks corresponding to the $\gamma - \gamma$ cascade. Since the energies of the associated emitted photons are 1173.2 keV and 1332.5 keV, these peaks seemed good candidates.

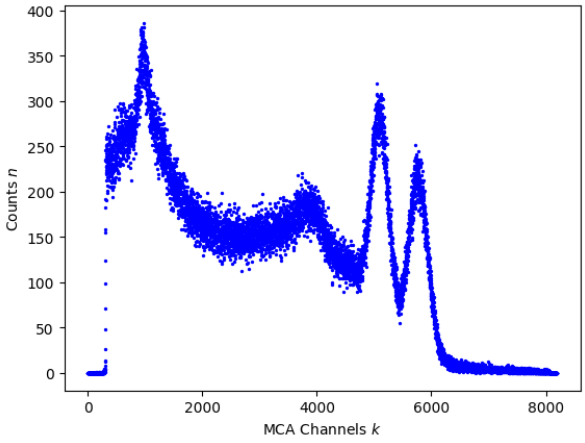


Figure 10: MCA plot of the spectrum of the analog signal from the stationary detector, after increasing the amplification of the main amplifier.

5.2 *Calibrating the SCAs*

We next introduced the SCAs into the system. After going through the amplifier, we divided the signal from the detectors into two, sending one through the SCA and then a GDG, while the other was sent through a microsecond delay generator. Both signals were then sent to the oscilloscope and the MCA (see Fig. 11), with the latter having the GDG signal set as its gate (the 'Gate' setting now being set to 'Coincidence').

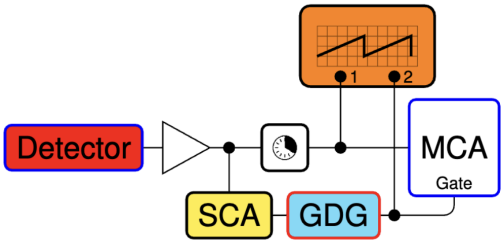


Figure 11: Circuit diagram for calibrating the SCAs. The box with a clock represents a microsecond delay generator. [5]

Using the oscilloscope, we adjusted the microsecond delay such that the analog signal peaks were in between the square wave gate from the GDG (Figs. 12 and 13). Thus, the plot on the MCA showed a cut version of the original spectrum, corresponding to the thresholds set on the SCA. We adjusted these thresholds until only the two photopeaks were fully visible in the energy window (see Figs. 14 and 15). For the moving detector, these thresholds were 6.71 V to 8.9 V, while for the stationary detector, they were 3.48 V to 7.70 V. Calculating the sensitivity

$$\eta = \frac{N_{\text{SCA-gated}(E)}}{N_{\text{Self-gated}(E)}}$$

and plotting this too with the spectra shows us that the sensitivity is approximately 100% around the photopeaks.



Figure 12: Analog signal from the moving detector on the oscilloscope (pink) with the digital signal from the SCA (yellow); the two coincide



Figure 13: Analog signal from the stationary detector on the oscilloscope (pink) with the digital signal from the SCA (yellow); the two coincide

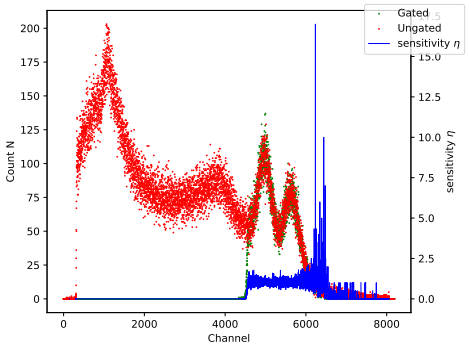


Figure 14: MCA plot of the spectra of the moving detector's signal, showing the full ungated spectrum, the gated (by SCA) cut spectrum, and the sensitivity.

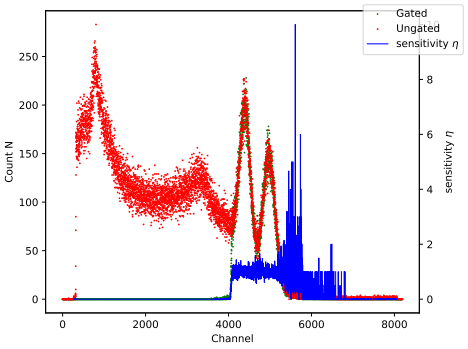


Figure 15: MCA plot of the spectra of the stationary detector's signal, showing the full ungated spectrum, the gated (by SCA) cut spectrum, and the sensitivity.

5.3 *Calibrating the CFDs*

After this, we introduced the CFDs to our circuit, splitting the original signal into two, one going via an amplifier to a microsecond delay generator, and another going via an inverse amplifier to a CFD and then a GDG. Both signals were then sent to an oscilloscope and MCA, with the MCA again using the latter signal as a gate (Fig. 16).

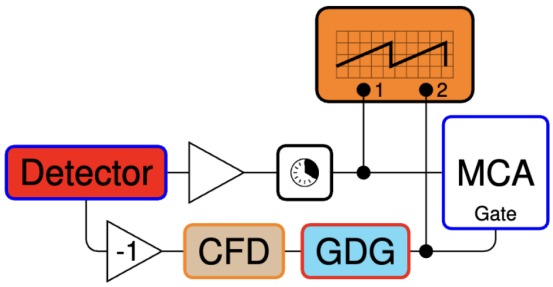


Figure 16: Circuit diagram for calibrating the CFDs. The triangle with a -1 in it represents the inverse amplifier. [5]

Again, as with the SCAs, we adjusted the delay until the analog part of the signal sat within the square gate (Figs. 17 and 18). We also adjusted the thresholds of the CFDs until only the two photopeaks from the original spectrum remained on the MCA (see Figs. 19 and 20). Calculating the sensitivity

$$\eta = \frac{N_{\text{SCA-gated}(E)}}{N_{\text{Self-gated}(E)}}$$

and plotting this too with the spectra shows us that the sensitivity is approximately 100% around the photopeaks.



Figure 17: Analog signal from the moving detector on the oscilloscope (pink) with the digital signal from the CFD (yellow); the two coincide



Figure 18: Analog signal from the stationary detector on the oscilloscope (pink) with the digital signal from the CFD (yellow); the two coincide

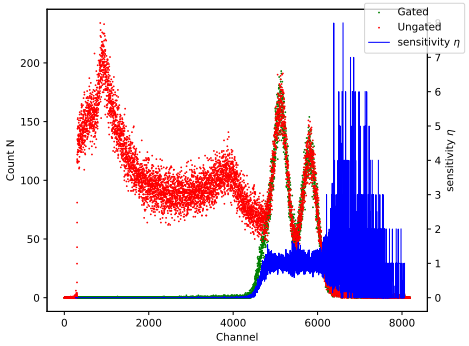


Figure 19: MCA plot of the spectra of the moving detector's signal, showing the full ungated spectrum, the gated (by CFD) cut spectrum, and the sensitivity.

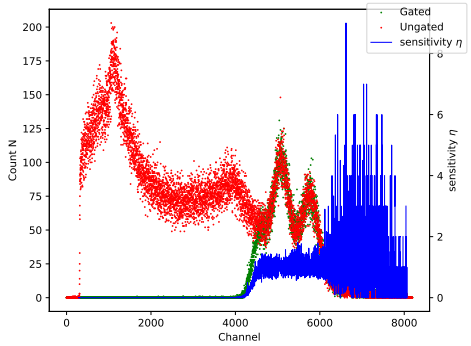


Figure 20: MCA plot of the spectra of the stationary detector's signal, showing the full ungated spectrum, the gated (by CFD) cut spectrum, and the sensitivity.

5.4 *Adjusting the Delay in the Fast Branch*

So far, we had been doing the calibrations for each detector one at a time. We now brought them both into the circuit, sending their signals via inverse amplifiers to CFDs and then to nanosecond delay generators. From here, both signals were split, with one part being sent to the oscilloscope and the other being sent through the fast-coincidence to counter 4 of the MCA (Fig. 21).

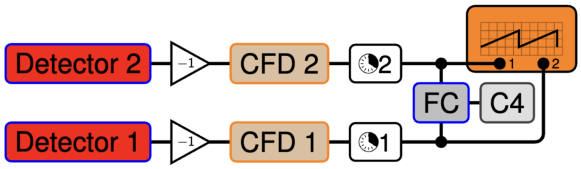


Figure 21: Circuit Diagram for adjusting the fast branch delay. The two delay generators here are nanosecond generators, the box labelled 'FC' is a fast-coincidence, and the box labelled 'C4' is Counter 4 of the MCA into which the signal is sent. [5]

We set the delay of the moving detector to be fixed at 32 ns and varied the delay of the stationary detector in the range of 1 ns to 63 ns in integer steps (i.e. an overall delay between the two signals of −31 to 31 ns). For each value of the difference in delays, we measured the number of counts recorded by the MCA in 10 seconds.

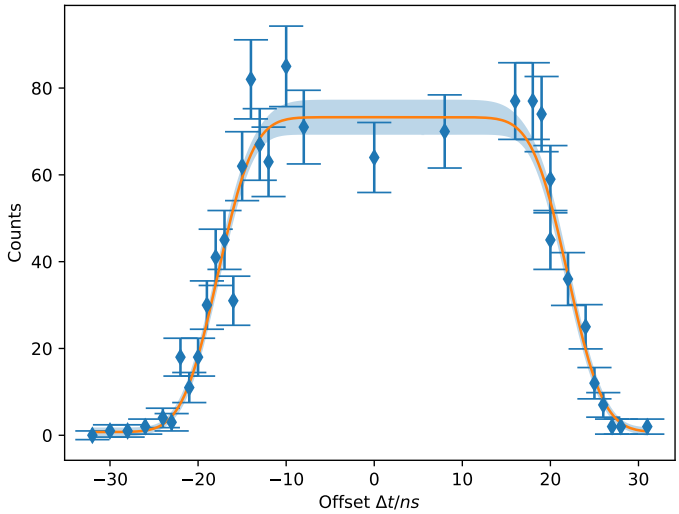


Figure 22: Prompt Curve from the Fast Branch, with experimental data and uncertainties in blue, and fit in yellow

The offset v/s counts data should follow the function

$$f(t) = \frac{A}{2} \cdot \left(1 + \text{erf}\left(\frac{t - (t_0 - \frac{w}{2})}{\sigma}\right) \cdot \text{erf}\left(\frac{(t_0 + \frac{w}{2}) - t}{\sigma}\right)\right) + A_0$$

where t is the time offset, A is a scaling factor, A_0 is the vertical offset on the plot, σ is the width of the slopes, w is the resolving time which controls the width of the

Parameter	Fit Value
A	72.3 ± 3.7
A_0	0.66 ± 0.61
σ	$4.44 \pm 0.42\,\mathrm{s}$
w	$39.71 \pm 0.68\,\mathrm{s}$
t_0	$2.07 \pm 0.18\,\mathrm{s}$

Table 1: Values of the Parameters as calculated by the fit

function, and t_0 is the centre of the plateau of the function. The erf function is the Gauss error function

$$\mathrm{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-t^2} dt$$

Changing the resolving time changes the width of the plateau, shortening or lengthening it as the resolving time decreases or increases. Having too short a resolving time would not be advantageous as there would only be a small window of delay offsets where we could expect to see a significant number of true coincidences. Having too long a resolving time would also be disadvantageous as now the window where we can measure random coincidences is too small. As can be seen from our plot above, the resolving time we had was just right for the purposes of this experiment. The values of the various parameters from doing the fit are as follows:

5.5 *Adjusting the Delay in the Slow Branch*

As the last step of the calibration before taking the main measurements, we brought in the slow branch to our circuit too (Fig. 23). We adjusted the microsecond delays and the GDG delay, along with the widths of the square wave gates to ensure that all three coincided, aligning their edges.

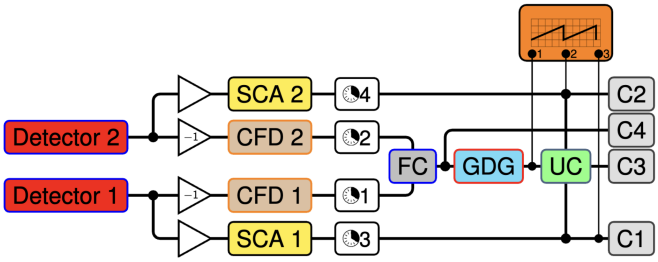


Figure 23: Circuit Diagram for adjusting the slow branch delay. The delay generators labelled 1 and 2 are nanosecond delay generators, while those labelled 3 and 4 are microsecond delay generators. The box labelled 'UC' is the universal coincidence, and the boxes labelled 'C1' to 'C4' are Counters 1 to 4 of the MCA. [5]



Figure 24: Digital signals from the slow branch on the oscilloscope, with the fast branch gate in yellow, the SCA gate from the stationary detector in pink, and the SCA gate from the moving detector in blue.

6 Analysis of the Data

6.1 *Calibration of the SCAs*

7 Main Analysis

With the constructed circuit the random measurement and main measurement are taken. For the random measurements a total of 13 angles ranging from 90° to 270° in 15° steps are taken. At each angle the count coincidence rate is measured for 10 Minutes. Since the random measurement is done in 2 runs, every angles was measured twice. For the main measurement the total runs were increased from 2 to 6 and the measuring time per angles per run was increased from 10 Minutes to 12 Minutes.

7.1 *Random Measurements*

7.1.1 *Direct Measurement*

To investigate the random coincidences of the signals, the delay in the fast branch is set to maximal difference to ensure no true coincidence is at play. Now only accidental overlapping signals trigger a coincidence. The rate of random coincidences can be determined by two methods. First, the gated counts are taken. Those already are the random coincidences since only when the signals of the fast branches overlap, a signal from the slow branch is received. As for all measured counts from the spectrum, a Poisson distribution is assumed, such that the uncertainty is

$$\sigma_N = \sqrt{N} \tag{5}$$

Suppose a count of 0 is registered. That would have yield an uncertainty of also 0. Since an uncertainty of zero is not realistic, a conservative correction is taken such

that an uncertainty of

$$\sigma_N = \sqrt{N+1} \tag{6}$$

is assumed. This yields the count rate distribution shown in Figure 25

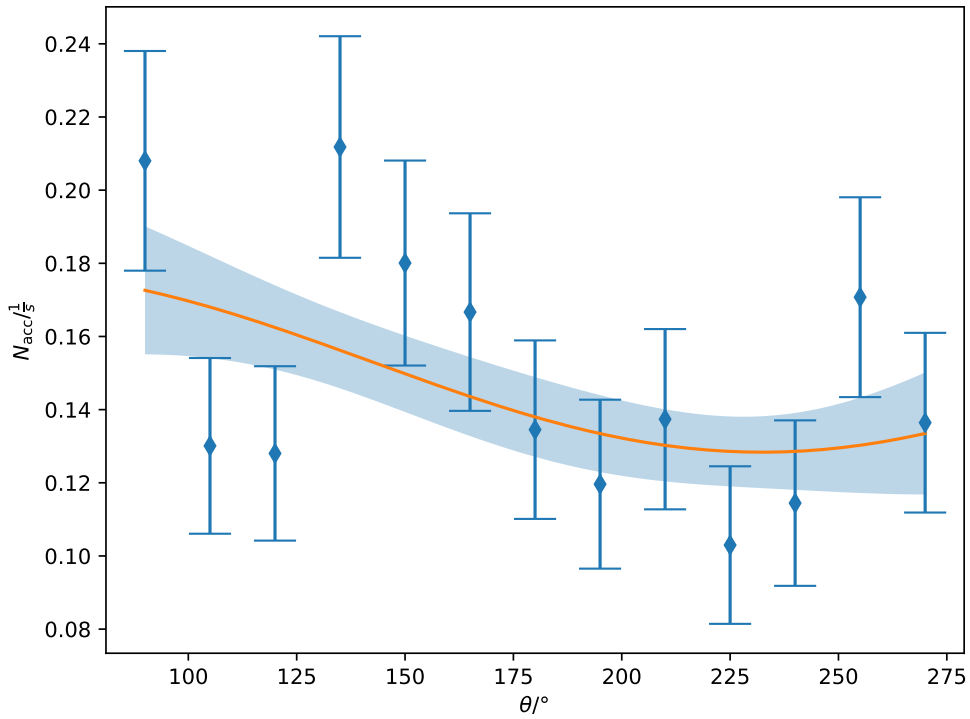


Figure 25: Direct Measurement of Random Coincidences from

It can be seen that the distribution is anisotropic, i.e. specific directions increase the count-rate. This can be caused by strong radio active sources in the vicinity of the experiment apparatus. Therefore, the rate of random coincidences is dependent on the angle. The shown fit is of the form

$$N(\theta) = a + b \cdot \cos\left(2 \cdot \pi \frac{\theta}{360} - c\right) \tag{7}$$

wtih paramters $a = 0.153(14), b = -0.025(15)$ and $c = 110.87(70)$ and has a quality of $\chi^2/\text{ddof} = 1.57$, proving the function to be valid for modeling the distribution. Here it should be notice that 2π periodicity is enforced, since for a full rotation of the moving detector, the same signal is expected.

7.1.2 *Resolving Time and Single Detector Count Rates*

Another method is to calculate the accidental coincident rate with the resolving time and the count rate of the two detectors. For this the formula

$$N_{\text{acc}} = \sigma \cdot N_1 \cdot N_2 [1] \tag{8}$$

is used. Here we have the count rates N and the resolving time $\sigma = 2\tau$. The resolving time is set at the fast coincidence module to $\sigma = 30(4)$ ns. The large uncertainty does not come from used modules, but represents the range of the resolving time such that the coincidence is reasonable.

The count rates are shown in Figures 26 to 27

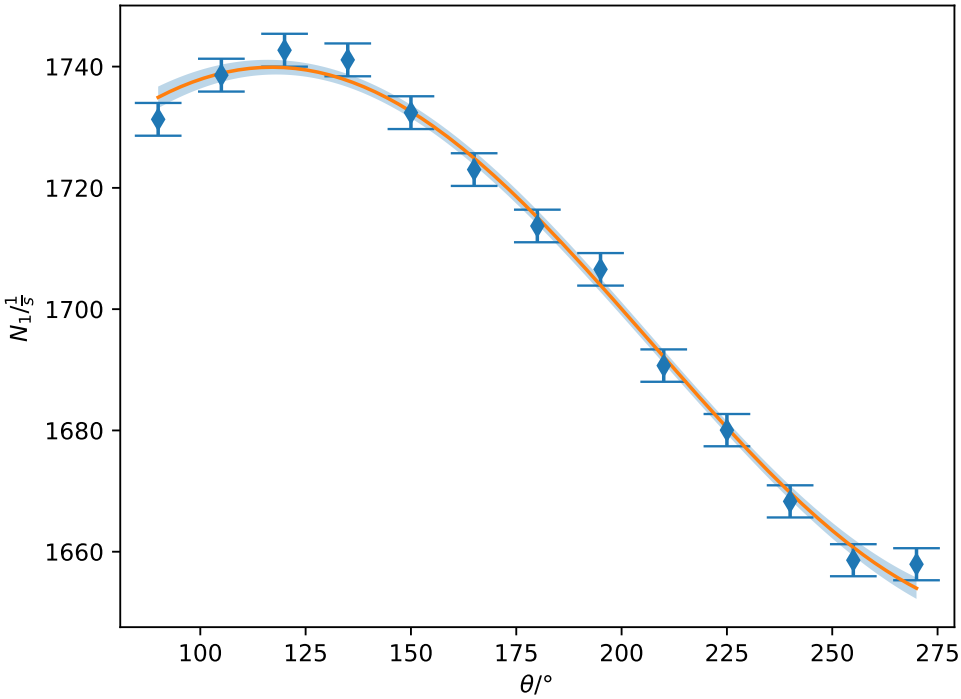


Figure 26: Count Rate for Detector 1 (Moving)

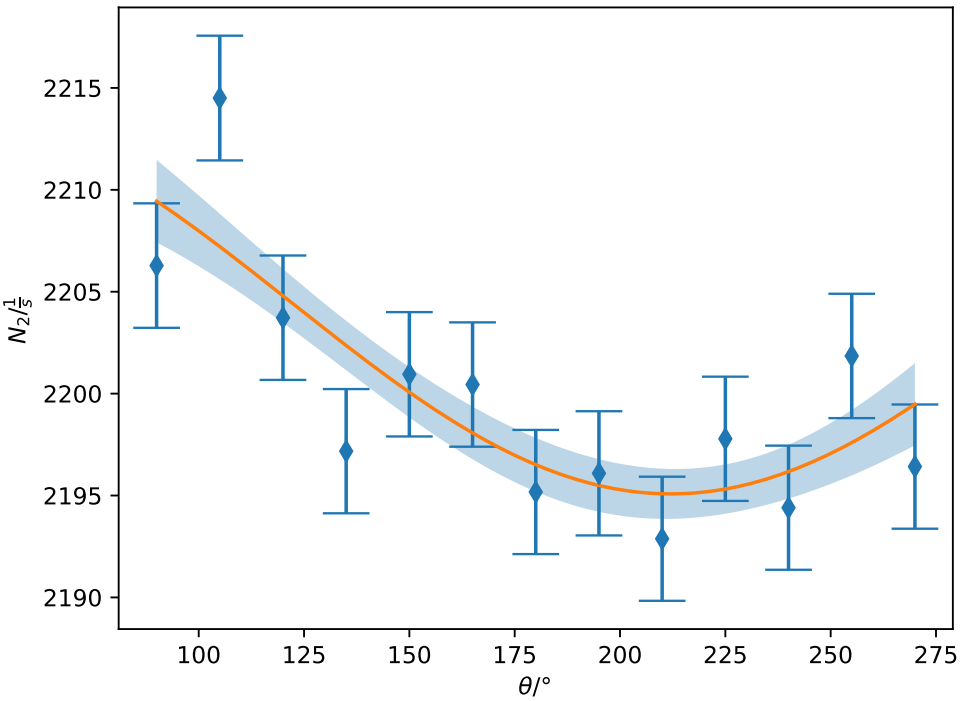


Figure 27: Count Rate for Detector 2 (Stationary)

As seen before in the direct measurement, an anisotropic distribution is observed in the moving detector. The quality $\chi^2/\text{ddof} = 0.97$ is well within a reasonable range with parameters $a = 1694.4(15), b = 45.5(13)$ and $c = 2.043(43)$. Optimally the quality should be 1. On the other hand, the quality $\chi^2/\text{ddof} = 1.52$ of the stationary detector is fairly high but also within a reasonable range wiht paramters $a = 2204.5(17), b = 9.4(22)$ and $c = 6.84(17)$. Even though the stationary detector should not be dependent on the angle, a dependency is still assumed, since the moving detector can screen of certain parts of the field of view of the stationary detector and thus influence the count rate. Here is also a 2π periodicity assumed. Using the Equation (8) again is probably not perfectly modeling the behaviour. A more appropriate Ansatz is an expansion in the spherical harmonics, where the data is fitted for the coefficients, since the spherical harmonics constitute a complete set. Taking another look at Equation (8) one might notice, that this is in fact already a expansion to first order in l and $m = 0$ of the spherical harmonics. This and the fact that the quality is close to 1, justifies still using Equation (8).

As a result of the count rates and resolution Figure 28 is obtained.

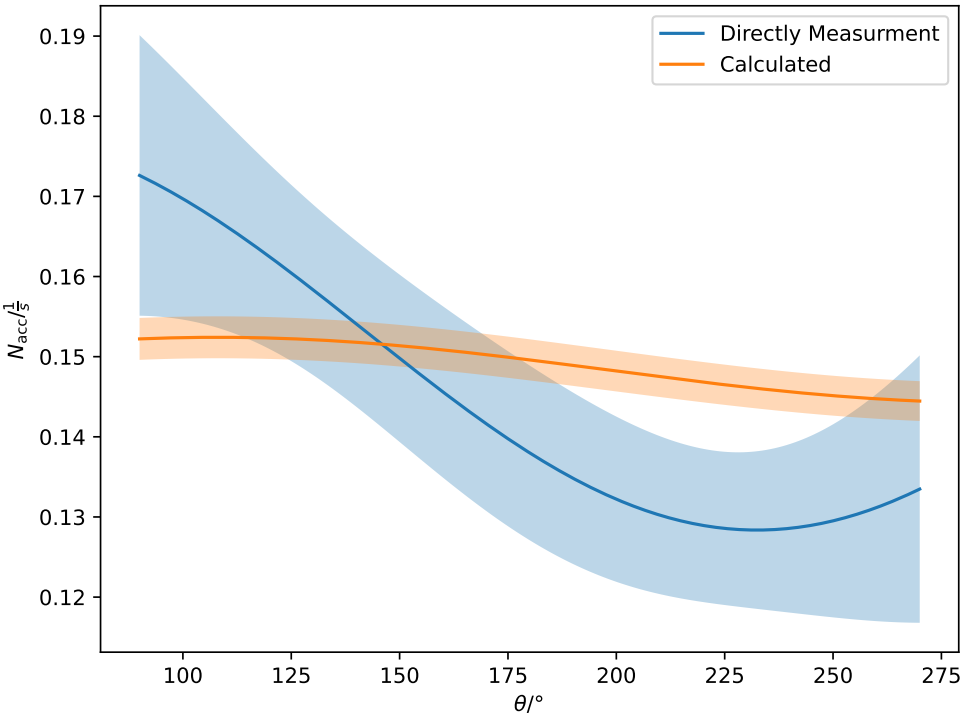


Figure 28: Accidental Coincidences Calculated with Resolving Time and Count Rates of Detector 1 and 2 and Accidental Coincidences from the Direct Measurement.

For this the functions $N_1(\theta)$ and $N_2(\theta)$ are used to calculate the accidental coincidence rate from Equation (8). Here the resolving time 39.71(68) ns from the prompt curve is used. Also the fitted function from the direct measurement from Figure 25 is shown here for comparison. As seen a small dependency on the angle is present for the calculated accidental coincidences. This behaviour is though more significant in the direct measurement. Both curves agree well with each other with deviations of only up to 2σ at an angle of about 225°. Though the agreement is good, for further calculations only the direct measurement is used since it includes also noise and other influences from the used devices.

7.2 *Stability Check*

To investigate the stability of the detectors over time, the main measurement is used to look at the fluctuations of the signals of the moving and stationary detector for constant angles over the multiple measurements for the different runs. Here fluctuations are defined as the standard deviation of of the signal over the different runs, since the runs happen at different points in time.

If the detectors would be stable over time, all fluctuations of the count rate would be within the uncertainty given by the Poisson distribution, i.e. the fluctuations could

be caused by the uncertainty. Therefor at each angle the standard deviation and the uncertainty is calculated. Here the uncertainty is calculated as for every weighted mean like

$$\bar{\sigma}_x = \sqrt{\frac{1}{\sum_i \sigma_i^{-2}}} \tag{9}$$

To compare the different angles to each other, the standard deviation and the uncertainty are then divided by the mean of the count rates for each angle. This of course makes them only approximative comparable, but that should suffice for this analysis. As a result Figure 29 and Figure 30 are obtained for the moving detector and stationary detector respectively.

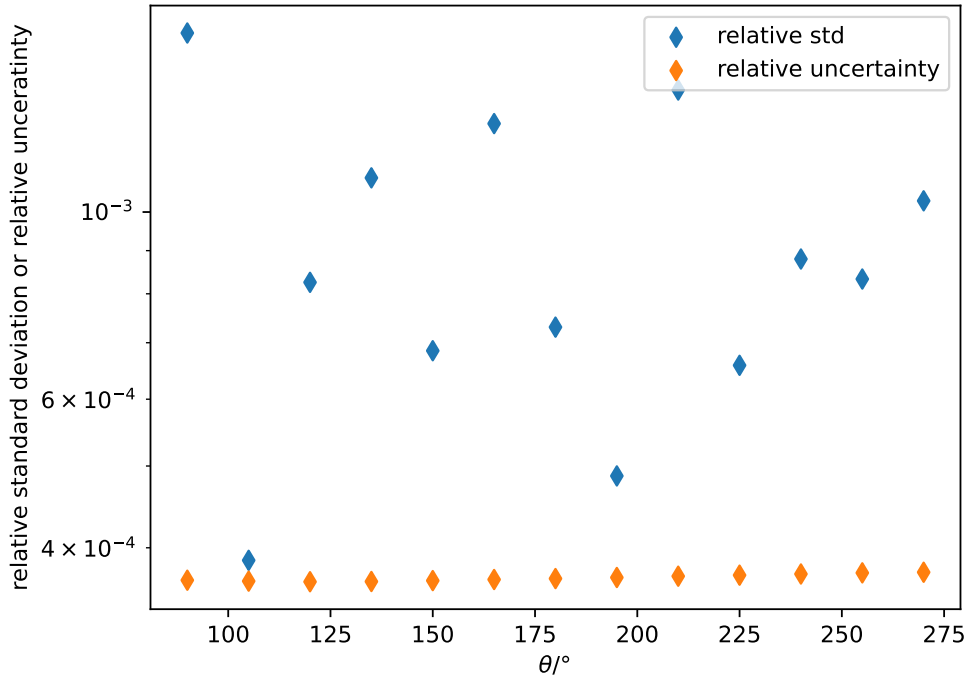


Figure 29: Relative Standard Deviation and Uncertainties of Count Rates for Constant Angles of the Moving Detector

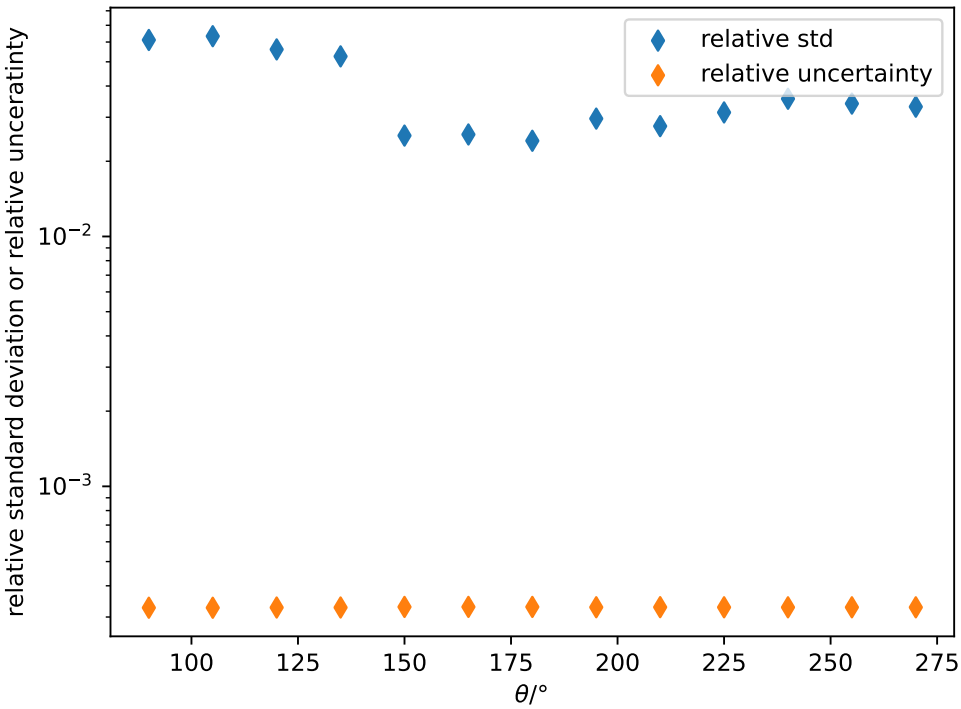


Figure 30: Relative Standard Deviation and Uncertainties of Count Rates for Constant Angles of the Stationary Detector

For the moving detector and all angles, the fluctuation is higher then the uncertainty. The detector is not stable over time and the counts depend on the time measured. The same is the case for the stationary detector. Therefore the count rate is dependent on the specific time. To get more insight into the behaviour of the time dependency, the count rates are shown in Figures 31 to 32. Here, over all runs are averaged. As always the weighted average is take, where the weight is the square of the inverse uncertainty.

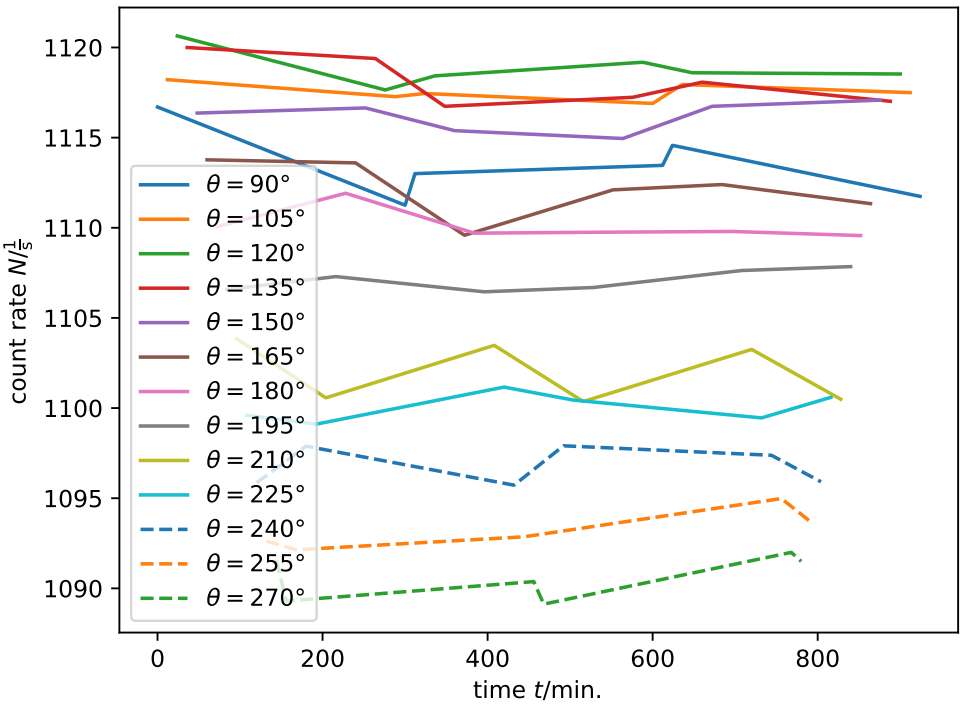


Figure 31: Count Rates for Constant Angles of Moving Detector over Time

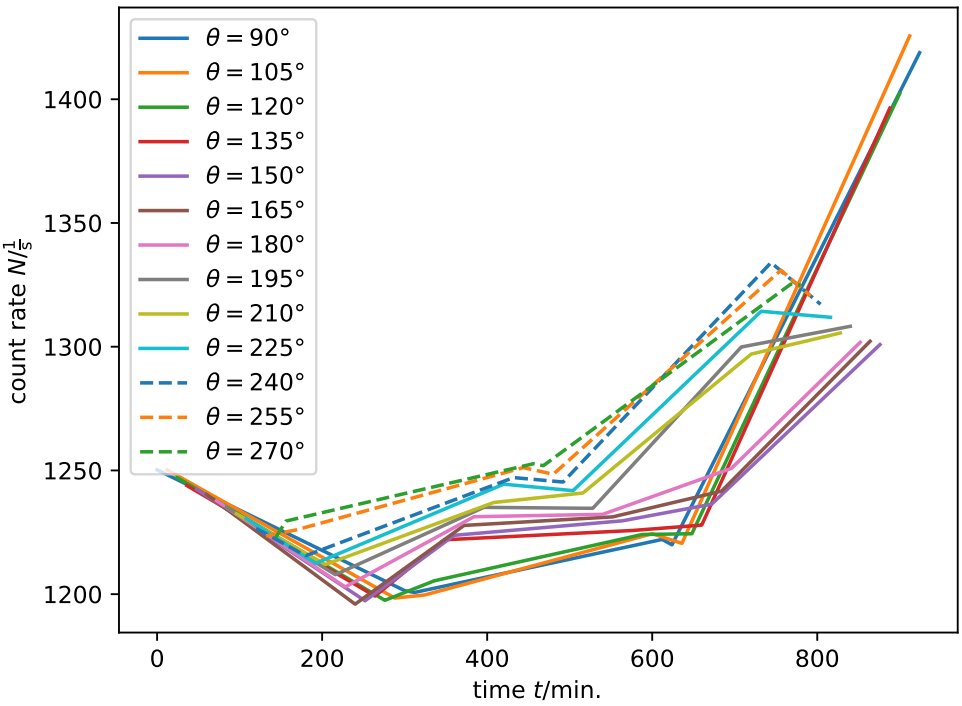


Figure 32: Count Rates for Constant Angles of Stationary Detector over Time

Against the usual manner, the data points are connected and uncertainties omitted. The purpose of this, is to enhance visibility for all 13 different angles, since only their general behaviour is of interest and not their exact values. From the figures themself it gets apparent that the count rate increases for all angles for the stationary detector, while this behaviour is even stronger for the first four angles. The moving detector has no such strong dependency on time.

The same procedure is now done with the total coincidence rate and with the fast branch coincidence rate instead of separating into detector 1 and 2, i.e. moving and stationary.

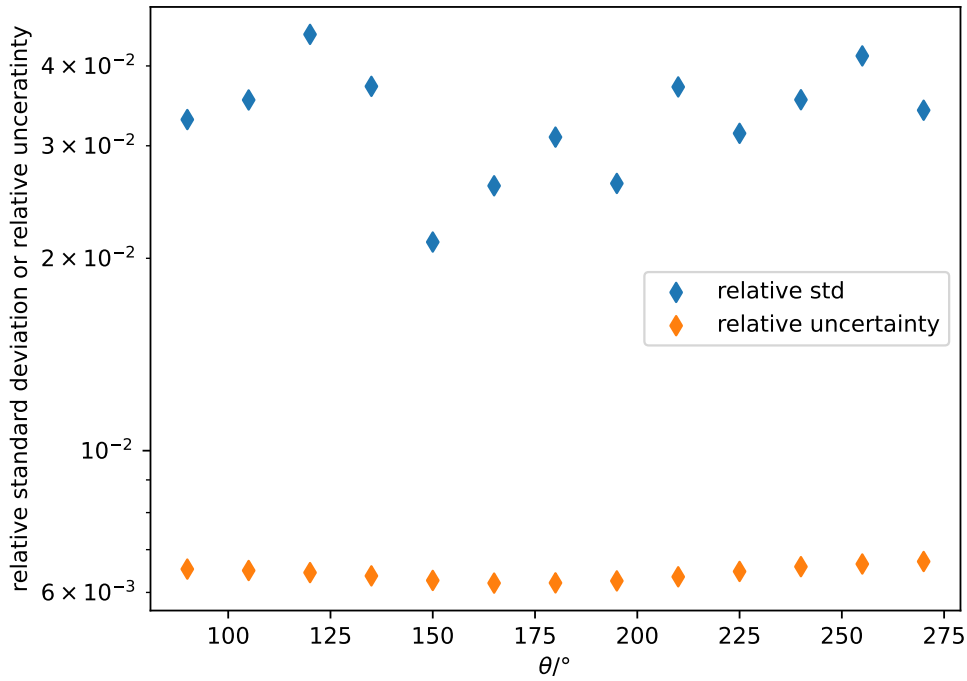


Figure 33: Relative Standard Deviation of Count Rates for Constant Angles for the Coincidence Rate

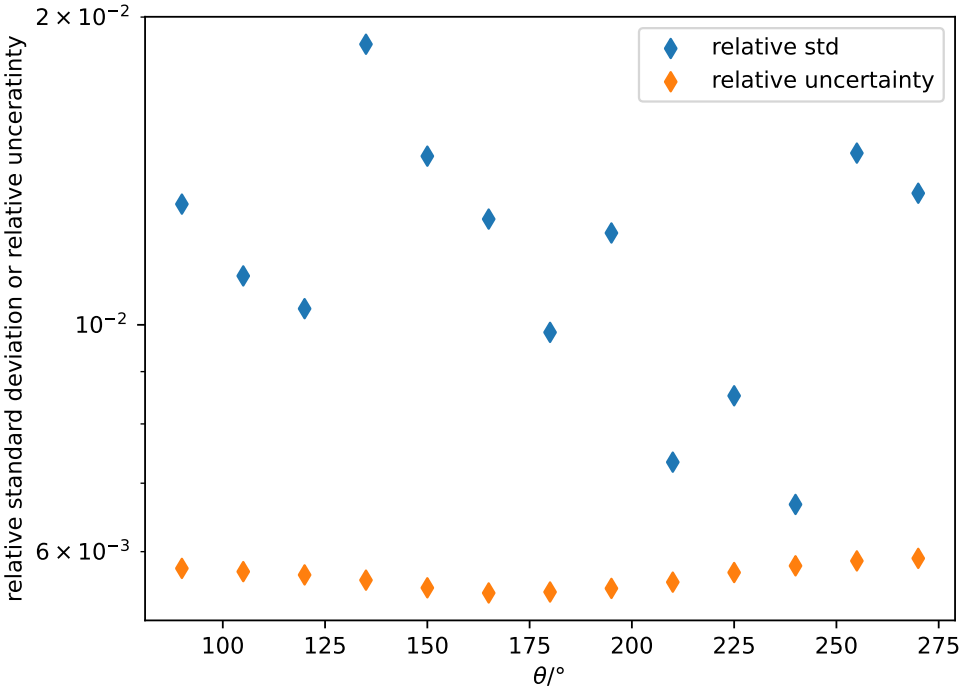


Figure 34: Relative Standard Deviation of Count Rates for Constant Angles of the Fast Branch Coincidence Rate

All fluctuations are above the uncertainty (see Figures 33 to 34), making them significant, i.e. non-neglectable. The count rates per angle over the measuring time are shown in Figures 35 to 36.

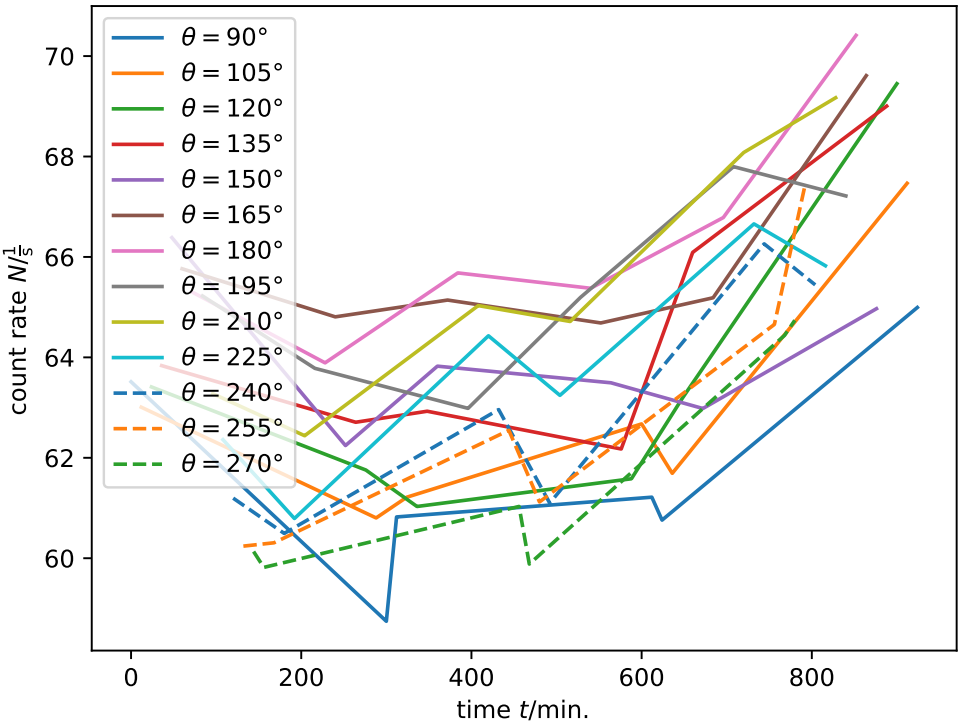


Figure 35: Coincidence Rate for Constant Angles over Time

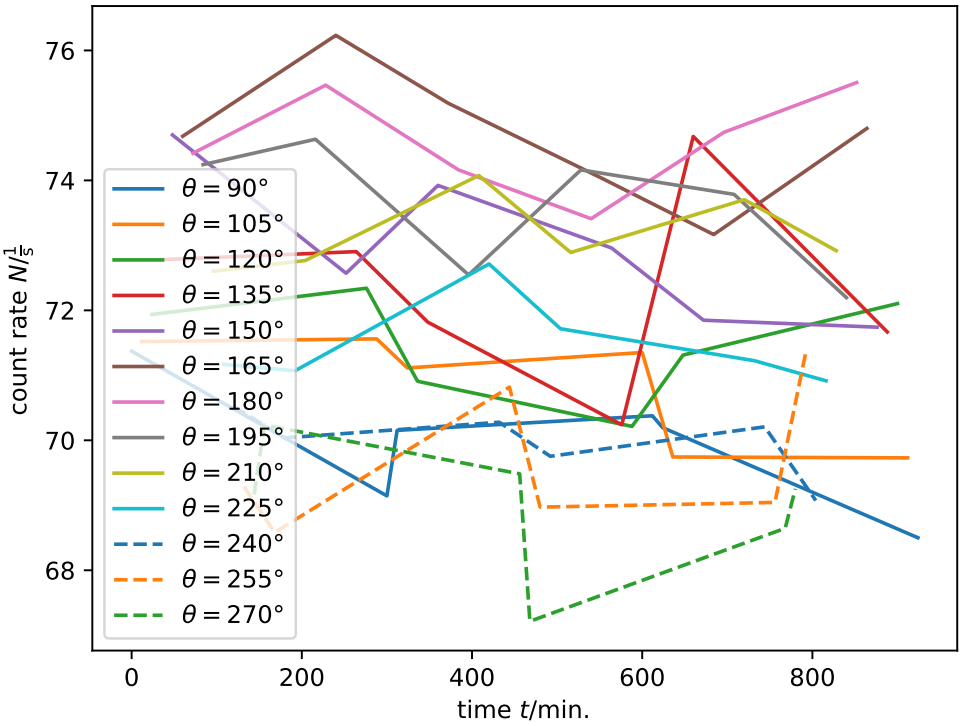


Figure 36: Fast Branch Coincidence Rate for Constant Angles over Time

The coincidence rate has a minimum around 300 minutes into the measurement for all angles, but increases again until the end to even higher coincidence rates then at the beginning of the measurement. The fast branch has no particular behaviour for the coincidence rates.

Overall, the time stability is poorly since the count rate is always dependent on the time when measured. Albeit this, the final result should not depend on this time dependency too strongly, since multiple measurements of the same angle are taken at different times, thereby should even out that time dependency.

To improve the measurement, it is advisable to make more runs and measure for less time per angle per run. For this measurement each angle was measured 12 minutes per run for a total of 6 runs and thus 936 minutes or 15 hours and 36 minutes. Doing e.g. 3 minutes per run for total of 24 runs would reduce the time dependency significantly, though taking the same amount. Of course the moving detector contributes a few extra seconds for repositioning but this should be neglectable. During the experiment the detector was noticed to take 2 to 4 seconds for moving.

7.3 *Correction for Random Coincidences*

From here on the random coincidences from the direct measurement in Section 7.1.1 are subtracted from the main measurement. The distinguishment between different angles is kept in case angular dependent photon sources skew the measurement.

7.4 *Misalignment Correction*

Considering that the source is not perfectly centered, its misalignment is investigated. For this an angle dependent correction factor

$$\alpha(\theta) = \frac{N(\theta)}{N(180°)} \tag{10}$$

with

$$N(\theta; r_d, r_s, \theta_s) = \frac{N_0}{\Delta R(r_d, r_s, \theta, \theta_s)^2}, \tag{11}$$

$$\Delta R(r_d, r_s, \theta, \theta_s) = (\vec{R}(r_d, \theta) - \vec{R}(r_s, \theta_s))^2 \tag{12}$$

$$\vec{R}(r, \theta) = r \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} \tag{13}$$

results in

$$\alpha(\theta) = \frac{r_d^2 - 2r_d r_s \cos(180° - \theta_s) + r_s^2}{r_d^2 - 2r_d r_s \cos(\theta - \theta_s) + r_s^2} \tag{14}$$

The main measurement data is used to show this ratio α in Figure 37

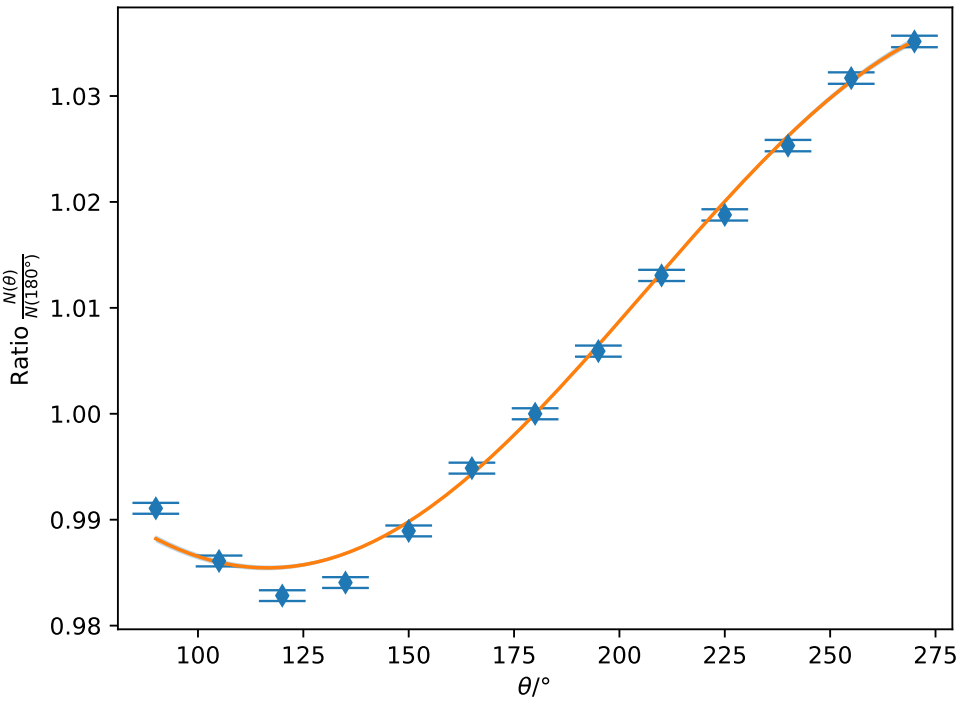


Figure 37: Misalignment Measurement

As a result the displacement $r_s = 64.9(13)\,\mu$m at angle $\theta_s = 116.5(14)°$ with a fit quality of $\chi^2/\text{ddof} = 9.82$ is found. The χ^2 is very high and shows that during the experiment and analysis larger systematic uncertainties have not been accounted for or wrong assumptions have been made.

This correction ratio $\alpha(\theta)$ can now be applied for the final analysis by multiplying the ratio with the count rates. It will correct of up to 3.5% as seen in Figure 37.

7.5 *Solid Angle Corrections*

Since the detector is not point-like but has a spacial volume, the incoming rays are not perfectly normal to the detector surface. Therefore a correction has to be made to coefficients

$$A_{ii} = \frac{A_{ii}^{\text{exp}}}{Q_{ii}(r_1)Q_{ii}(r_2)} \tag{15}$$

of the expansion of the angular correlation function

$$W(\theta) = N_0 \cdot (1 + A_{22}P_2(\cos\theta) + A_{44}P_4(\cos\theta)) \tag{16}$$

Since the in the previous section determined displacement of 64.9(13) μm is within the uncertainty of 1 mm of the authors ability to measure the distances, the source can be seen as centered. Additionally assuming that the detectors have the same distance to the source, Equation (15) can be simplified with $r_1 = r_2 = r = \frac{d}{2}$ where $d = 90(1)$ mm is the distance from detector 1 to detector 2. Additionally it i assumed that the scintillator is 5 mm deep into the detector, making the total distance 10(1) mm. Therefore the detectors are separated by 5 cm from the source. Numerical values are given for the correction factors where they depend on this distance and the energy of the photon [4]. Even though the observed transitions happen at 1173.2 keV and 1332.5 keV, the correction factors plateau beyond 1 MeV. Thus the correction factors $Q_{22}(r) = 0.9057$ and $Q_{44}(r) = 0.7113$ for photons of 1 MeV are used.

7.6 *Angular Correlation Function*

The count rates are now corrected by misalignment and random coincidences by multiplying with the correction factor $\alpha(\theta)$ and subtracting the random coincidences

$$N_{\text{corr}}(\theta) = N(\theta) \cdot \alpha(\theta) - N_{\text{rand}}(\theta) \tag{17}$$

Here the random coincidence function $N_{\text{rand}}(\theta)$ corresponds to Equation (7) with the parameters from the direct measurement of the random coincidences. As a result the distribution in Figure 38 is obtained.

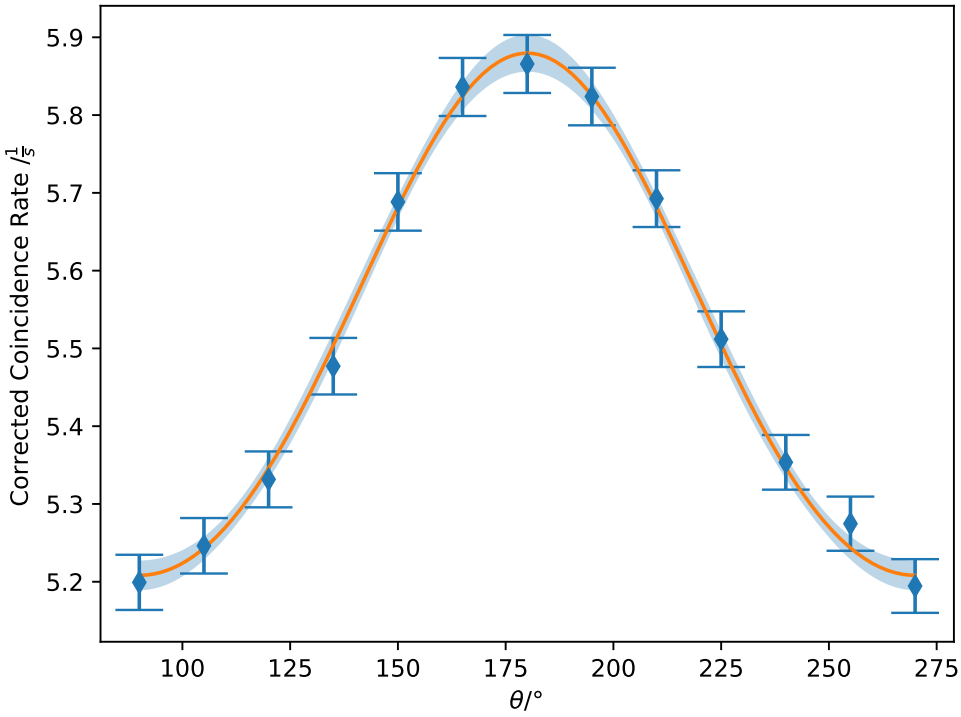


Figure 38: Main Measurement

Here the fit to the angular correlation function in Equation (16) is done. The quality $\chi^2/\text{ddof} = 0.22$ with parameters $N_0 = 5.411(11)$, $A_{22}^{\text{exp}} = 0.0799(37)$ and $A_{44}^{\text{exp}} = 0.0067(47)$ is achieved. The χ^2 is very small showing that at some point the uncertainties overestimated or plainly more data is needed to reduce the error. With the correction from the previous section the coefficients are adjusted to $A_{22} = 0.097(13)$ and $A_{44} = 0.0132(93)$. Given their large uncertainties, they lie within 1σ of the literature values $A_{22}^{\text{lit}} = 0.1020$ and $A_{44}^{\text{lit}} = 0.0091$ [4]. Especially $A_{44} = 0.0045(93)$ has an uncertainty larger then the value itself.

Last but not leas, other cascades are fitted to the experimental distribution. The chosen cascades and the fit results are in Figure 39 with Table 2. Here the coefficients $A_{ii} = F_i(L_1L_1I_iI) \cdot F_i(L_2L_2I_fI)$ for a $I_i(L_1)I(L_2)I_f$ cascade are set to literature values of the corresponding cascade[4] and only the normalisation N_0 is optimized.

cascade	χ^2/ddof	N_0
$0-1-0$	657.31	4.9661(92)
$0-2-0$	1647.61	4.4778(84)
$2-1-0$	12.22	5.4537(99)
$4-1-0$	255.81	5.553(10)
$4-2-0$	0.28	5.4087(98)

Table 2: Quality for Normalisation Constant N_0 Fitted to Different Cascades

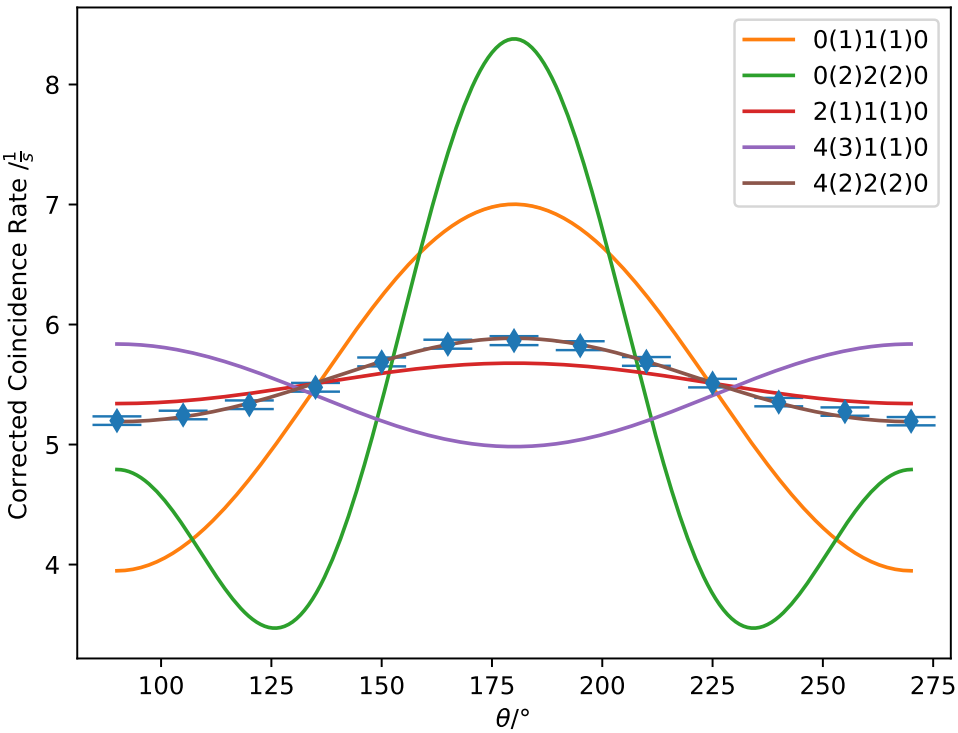


Figure 39: Different Cascades Fitted to the Experimental Angular Distribution

As seen, the assumed $4-2-0$ is most likely correct since the quality is here best, even though the χ^2 is quite low, which shows, that more data is needed.

8 Conclusion

In this experiment, we used a Fast-Slow coincidence setup with two detectors to measure the angular distribution of the radiation from the $\gamma-\gamma$ cascade of ^{60}Co, first by measuring random background coincidences, and then a main measurement of the true coincidences.

After collecting the data from the random and main measurement, the main analysis started. For this the random coincidence rate was determined by directly measuring the coincidence, and by calculating it with the single count rates of the moving and stationary detector and the resolving time.

Following the stability is measured by comparing the over-time-fluctuations with the uncertainty. There it is seen that the measurement is not stable over time. The influence on the date should still remain marginal though, each angle was measured at multiple time points.

Correcting for misalignment as been done by multiplying by a factor corresponding to an increment in count rate by being closer to the source for specific angles. Here it could also be determined that the source is off-center by 64.9(13) μm at an angle of 116.5(14)° from the stationary detector.

The last correction made is caused by the spacial volume of the used detector. Therefor the expansion coefficients in the angular correlation function have been corrected by a factor corresponding to a distance of 50 mm and already known from literature[4].

Finally, the angular correlation is plotted and fitted to the expansion. There coefficients $A_{22} = 0.097(13)$ and $A_{44} = 0.0132(93)$ are measured. They are within 1σ of the literature values $A_{22}^{\text{lit}} = 0.1020$ and $A_{44}^{\text{lit}} = 0.0091$ due to their large uncertainty. Also, other cascades have been considered. There it could bee seen that the presumed 4-2-0 cascade is most likely.

References

[1] Leo, W.R.: Techniques for Nuclear and Particle Physics Experiments: A How-to Approach, 2nd edn. Springer, Berlin (1994)

[2] Rosinsky, P.: Aufbau und Testung der Automatisierung Winkelabhängiger γ-Messungen Im Praktikum Mit Hilfe Einer Motorisierten Messapparatur

[3] Steffen, R.M.: Extranuclear effects on angular correlations of nuclear radiations. Advances in Physics **4**(15), 293 (1955) https://doi.org/10.1080/00018735500101234

[4] Siegbahn, K.: Alpha-, Beta-, and Gamma-Ray Spectroscopy vol. 2, p. 1695. North-Holland Publishing Company, Amsterdam (1965)

[5] Lab Handout